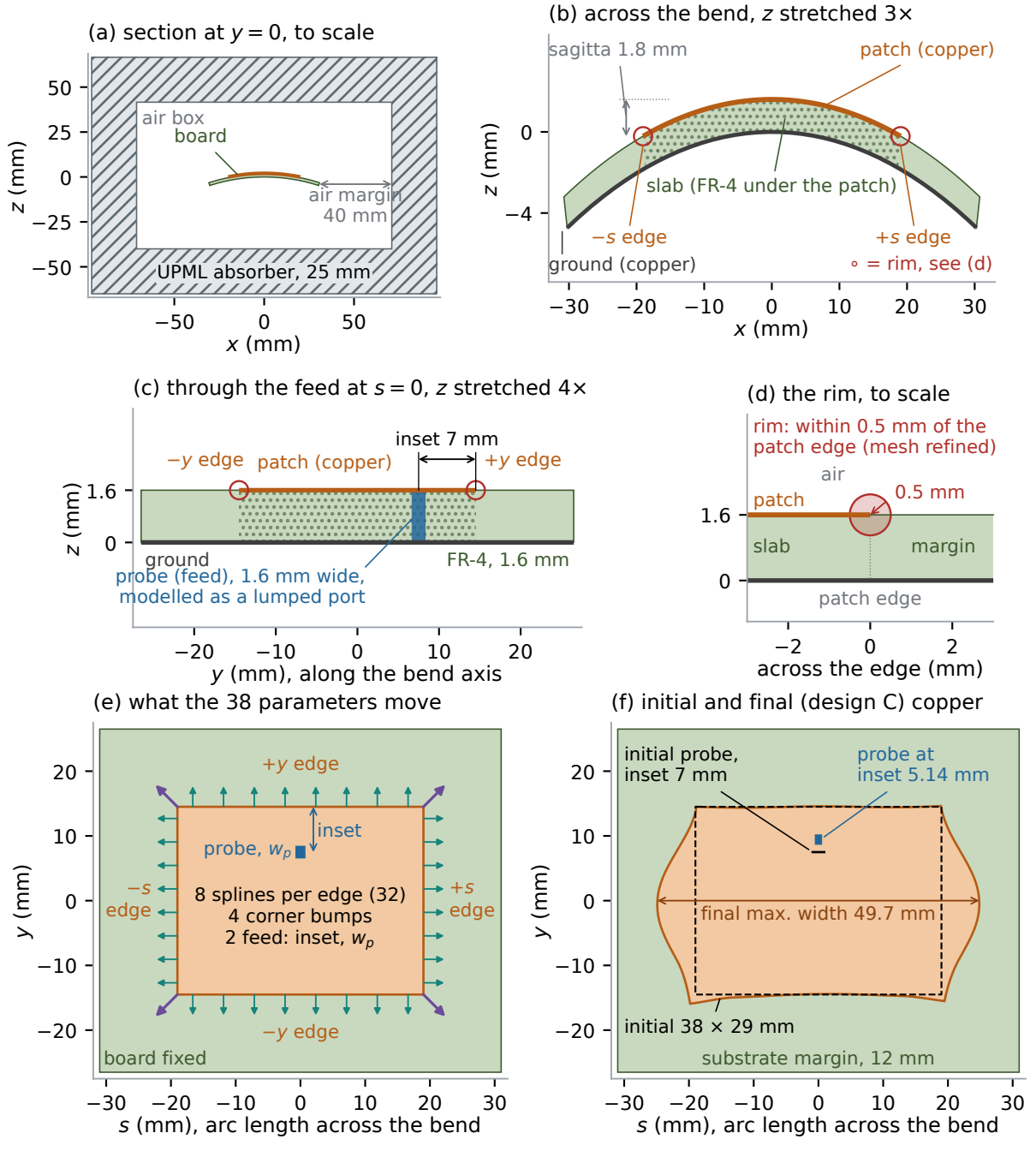

(a) section at $y = 0$, to scale
air box
board
air margin
40 mm
UPML absorber, 25 mm
z (mm)
x (mm)
(b) across the bend, z stretched 3×
sagitta 1.8 mm
patch (copper)
slab (FR-4 under the patch)
−s edge
+s edge
ground (copper)
◦ = rim, see (d)
z (mm)
x (mm)
(c) through the feed at $s = 0$, z stretched 4×
inset 7 mm
−y edge
patch (copper)
+y edge
ground
FR-4, 1.6 mm
probe (feed), 1.6 mm wide,
modelled as a lumped port
z (mm)
y (mm), along the bend axis
(d) the rim, to scale
rim: within 0.5 mm of the
patch edge (mesh refined)
air
patch
0.5 mm
slab
margin
patch edge
across the edge (mm)
(e) what the 38 parameters move
+y edge
inset
probe, w_p
−s edge
+s edge
8 splines per edge (32)
4 corner bumps
2 feed: inset, w_p
−y edge
board fixed
y (mm)
s (mm), arc length across the bend
(f) initial and final (design C) copper
initial probe,
inset 7 mm
probe at
inset 5.14 mm
final max. width 49.7 mm
initial 38 × 29 mm
substrate margin, 12 mm
y (mm)
s (mm), arc length across the bend